

**Exercise 2.5.** Prove the following.

- (a) If  $n$  is an integer, then  $n^2 + n$  is even.
- (b) If  $n$  is an integer, then  $3n^2 + 5n + 1$  is odd.
- (c) If  $n$  is an integer, then  $n^2 + 3n - 6$  is even.
- (d) If  $m$  and  $n$  are integers, then  $7m - 3n$  is even.

Solution.

We will use the definitions:

$$\text{Even integer} = 2k$$

$$\text{Odd integer} = 2k + 1$$

for some  $k \in \mathbb{Z}$ .

**(a) If  $n$  is an integer, then  $n^2 + n$  is even.**

**Proof**

Factor the expression:

$$n^2 + n = n(n + 1)$$

For any integer  $n$ , one of  $n$  and  $n + 1$  must be even because they are consecutive integers.

Therefore, their product must be even.

Hence,

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| $n^2 + n$ is even. |
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### Another way to prove it

We can consider the two possibilities.

Case 1:  $n$  is even.

Let

$$n = 2k.$$

Then

$$\begin{aligned}n^2 + n &= (2k)^2 + 2k \\&= 4k^2 + 2k \\&= 2(2k^2 + k),\end{aligned}$$

which is even.

Case 2:  $n$  is odd.

Let

$$n = 2k + 1.$$

Then

$$\begin{aligned}n^2 + n &= (2k + 1)^2 + (2k + 1) \\&= 4k^2 + 4k + 1 + 2k + 1 \\&= 4k^2 + 6k + 2 \\&= 2(2k^2 + 3k + 1),\end{aligned}$$

which is even.

Thus, in either case,

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| $n^2 + n$ is even. |
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**(b) If  $n$  is an integer, then  $3n^2 + 5n + 1$  is odd.**

We prove this by considering whether  $n$  is even or odd.

**Case 1:  $n$  is even**

Let

$$n = 2k.$$

Then

$$\begin{aligned} 3n^2 + 5n + 1 &= 3(2k)^2 + 5(2k) + 1 \\ &= 12k^2 + 10k + 1 \\ &= 2(6k^2 + 5k) + 1. \end{aligned}$$

This has the form  $2k + 1$ , so it is odd.

**Case 2:  $n$  is odd**

Let

$$n = 2k + 1.$$

Then

$$3n^2 + 5n + 1 = 3(2k + 1)^2 + 5(2k + 1) + 1.$$

Expand:

$$\begin{aligned} &= 3(4k^2 + 4k + 1) + 10k + 5 + 1 \\ &= 12k^2 + 12k + 3 + 10k + 6 \\ &= 12k^2 + 22k + 9 \\ &= 2(6k^2 + 11k + 4) + 1. \end{aligned}$$

Again, this has the form  $2k + 1$ , so it is odd.

Therefore, in both cases,

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| $3n^2 + 5n + 1$ is odd. |
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(c) If  $n$  is an integer, then  $n^2 + 3n - 6$  is even.

Factor the first two terms:

$$n^2 + 3n - 6 = n(n + 3) - 6.$$

Now  $n$  and  $n + 3$  have opposite parity:

- If  $n$  is even,  $n + 3$  is odd.
- If  $n$  is odd,  $n + 3$  is even.

Therefore,

$$n(n + 3)$$

is always even.

Also,

$$6$$

is even.

The difference of two even integers is even.

Therefore,

$$\boxed{n^2 + 3n - 6 \text{ is even.}}$$

**(d) If  $m$  and  $n$  are integers, then  $7m - 3n$  is even.**

🚩 As written, this statement is false.

Consider the counterexample

$$m = 1, \quad n = 0.$$

Both  $m$  and  $n$  are integers. But

$$7m - 3n = 7(1) - 3(0) = 7.$$

Since 7 is odd,

$7m - 3n$  is not necessarily even.

So there is no valid proof of (d) as written.

**What condition would make (d) true?**

For example, if the exercise said:

| If  $m$  and  $n$  are odd integers, then  $7m - 3n$  is even.

then it would be true.

Let

$$m = 2a + 1, \quad n = 2b + 1.$$

Then

$$\begin{aligned} 7m - 3n &= 7(2a + 1) - 3(2b + 1) \\ &= 14a + 7 - 6b - 3 \\ &= 14a - 6b + 4 \\ &= 2(7a - 3b + 2), \end{aligned}$$

which is even.